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ChE 311 Quiz 3
Spring 2019



Closed book and closed notes
5 minutes, 10 points

1. The flow defined by the following velocity field is incompressible flow.

$$\vec{v} = (t - x - y + z)\vec{\delta}_x + (2t + x + 2y + z)\vec{\delta}_y + (3t - x - y + z)\vec{\delta}_z \quad \nabla \cdot \vec{v} = 0$$

$$-1 + 2 + 1 = 2 \neq 0$$

A. True

B. False

Answer: B

2. The isothermal flow defined by the following velocity field in a cylindrical coordinate satisfies the continuity equation. The fluid is with constant density.

$$\vec{v} = (-r)\vec{\delta}_r + (2t + r + z)\vec{\delta}_\theta + (2z)\vec{\delta}_z$$

$$-1 + 0 + 2$$

A. True

B. False

Answer: B

3. Please write down Navier-Stokes function in vector format.

$$\rho \left(\frac{D\vec{v}}{Dt} \right) = \mu \nabla^2 \vec{v} - \nabla p + \rho \vec{g} \quad \text{or}$$

$$\rho \left(\frac{\partial \vec{v}}{\partial t} \right) =$$

$$-\nabla \cdot \vec{\tau} - \nabla p + \nabla (\rho \vec{v} \vec{v}) + \rho \vec{g}$$

4. Under which condition the initial term, $\rho \vec{v} \cdot \nabla \vec{v}$, in the N-S function can vanish as a reasonable approximation of the easier calculation.

(A) Low Reynolds number, creeping flow.

(B) High Reynolds number, inviscid flow.

(C) Unidirectional flow, rectilinear motion.

(D) Turbulent flow.

Answer: A

(A, C)

$$Re = \rho \langle v \rangle$$

5. Estimate the order of the Reynolds number of a swimmer who swims a kilometer in 20 minutes. Choose the closest answer.

(A) 0.01

(B) 1

(C) 100

(D) 10,000

Answer: B

$$\frac{1 \text{ km} \times \frac{1000}{1 \text{ km}}}{20 \text{ min} \times \frac{60 \text{ s}}{1 \text{ min}}} = \frac{10}{20 \times 60}$$

$$\frac{10}{1200} = \frac{1}{120} \approx 0.008$$